

# ANH TONG

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## EDUCATION

### University of Utah — Salt Lake City, UT

Aug 2021 – Dec 2025

B.S. in Data Science | GPA: 3.556 / 4.0 | Dean's List: Fall 2020, Spring 2024, Spring 2025

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## SKILLS

**Languages:** Python, SQL

**ML / DL:** PyTorch, TensorFlow, scikit-learn, XGBoost, HuggingFace (CLIP, Transformers)

**LLM / RAG:** LangChain, Ollama, OpenAI API, pgvector, Pinecone

**Data & Infra:** Spark, Feast, Redpanda/Kafka, PostgreSQL, Redis, Docker, MLflow, FastAPI, GitHub Actions

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## PROJECTS

### Apateur | Photography Settings & Lens Recommendation Engine

May – June 2026

- Upload a reference photo and your camera gear, Apateur finds visually similar real-photographer shots, returns the settings distributions they actually used (f-stops, ISO, speed), ranks your lenses for that shot type, and recommends what to buy next; grounded natural-language explanation generated via RAG (LangChain + Llama 3.2 / Ollama)
- Visual matching through CLIP (HuggingFace / PyTorch) embeddings stored in pgvector; zero-shot shot-type classification across 12 categories with no manual labels; over 22k Unsplash photos embedded and indexed via PostgreSQL.
- Lens ranker applies exposure-equivalence math to score user gear against the shot recipe; RAG layer (LangChain + Llama 3.2 / Ollama) uses strict generation rules to prevent hallucination, output is grounded in real EXIF data; containerized via Docker.

### ShopSentry | Real-Time Fraud Detection Platform

March – April 2026

- End-to-end ML platform for real-time anomaly detection using Redpanda, Spark Streaming, Feast feature store, MLflow experiment tracking, and Prometheus monitoring - full production ML lifecycle.
- Designed and compared 5 pseudo-labeling strategies (Isolation Forest, Autoencoder, Heuristics, Ensemble, Oracle) with stratified 5-fold CV; achieved F1 = 0.84, 100% bot and fraud recall.
- Deployed FastAPI prediction API with p50 = 32ms at 20 concurrent users, 0 failures over 30k+ load-tested requests.

### Cooked | AI-Powered Recipe Discovery App | Capstone

Jan – Dec 2025

- Led AI/data engineering on a team-built recipe app; designed semantic search over 3k+ recipes using transformer embeddings indexed in Pinecone, served via FastAPI.
- Built a grounded recipe chatbot that recommends dishes from ingredients on hand, responses are constrained to the live database with clickable results that route directly to full recipe pages.
- Enforced consistent recipe rendering across all navigation surfaces (home feed, saved, search results) so every entry point resolves to identical content and layout.
- Used LLM APIs (OpenAI, DeepSeek) to impute structured ingredient metadata (type, category, diet, nutrition) into a normalized schema across a 3,000-recipe Kaggle dataset.

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## WORK EXPERIENCE

### Salt Lake County Youth Services — Salt Lake City, UT

Dec 2024 – May 2026

Youth Leader, After School Program

- Designed and taught weekly data science curriculum for cohorts of 5–20 high school students, covering probability, programming, and financial literacy through hands-on simulations (blackjack, stock market, heat maps).
- Mentored students through personal challenges and post-graduation planning.
- Ran A/B experiments on content format (video vs. static) for the program Instagram, growing views ~9x in 30 days; applied the same iterative testing mindset used in ML model evaluation.

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## LEADERSHIP

### Data Science Club — University of Utah | Outreach & Social Media Lead

Oct – Dec 2025

- Grew event attendance from ~3 to ~30 participants in under 2 months by redesigning outreach strategy.